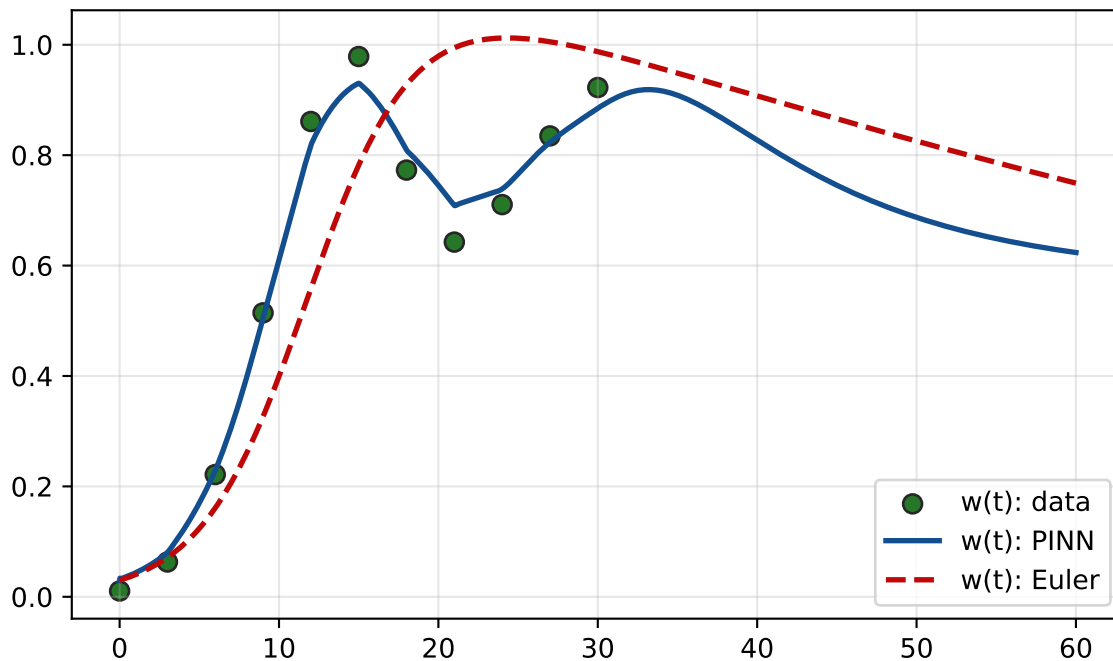
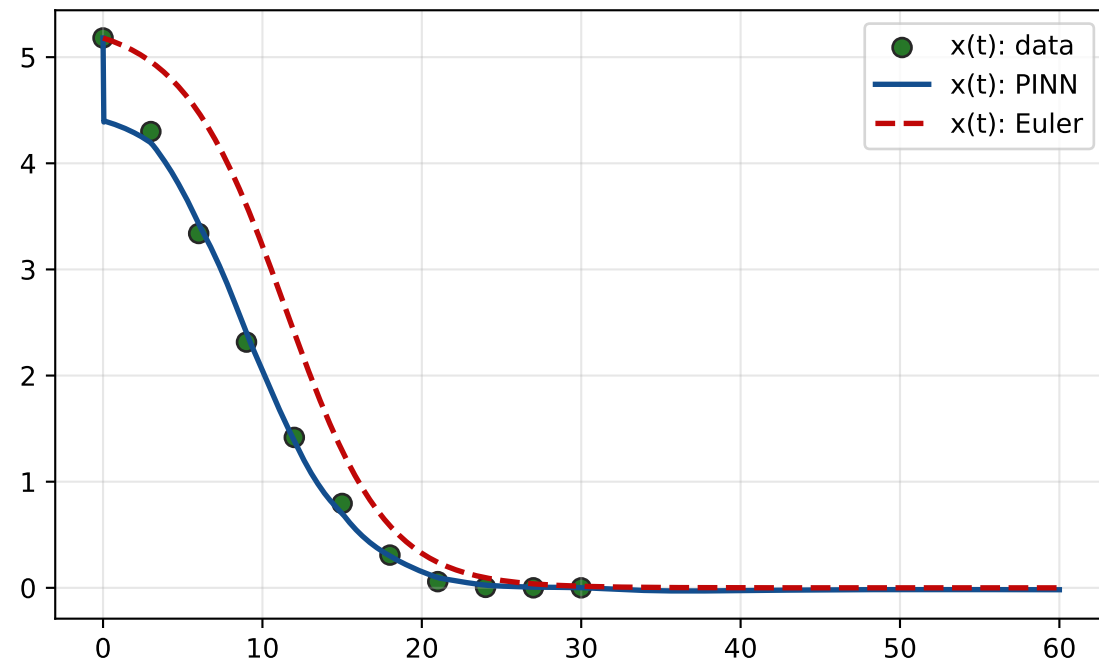


## E8 | PINN vs Euler

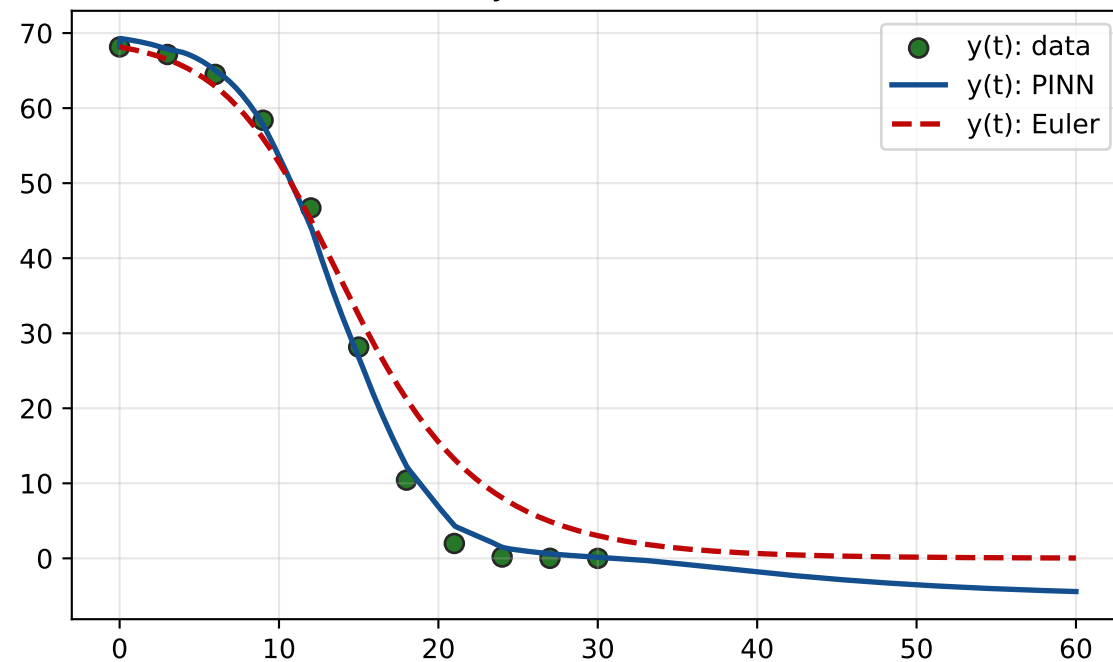
w(t): Biomass



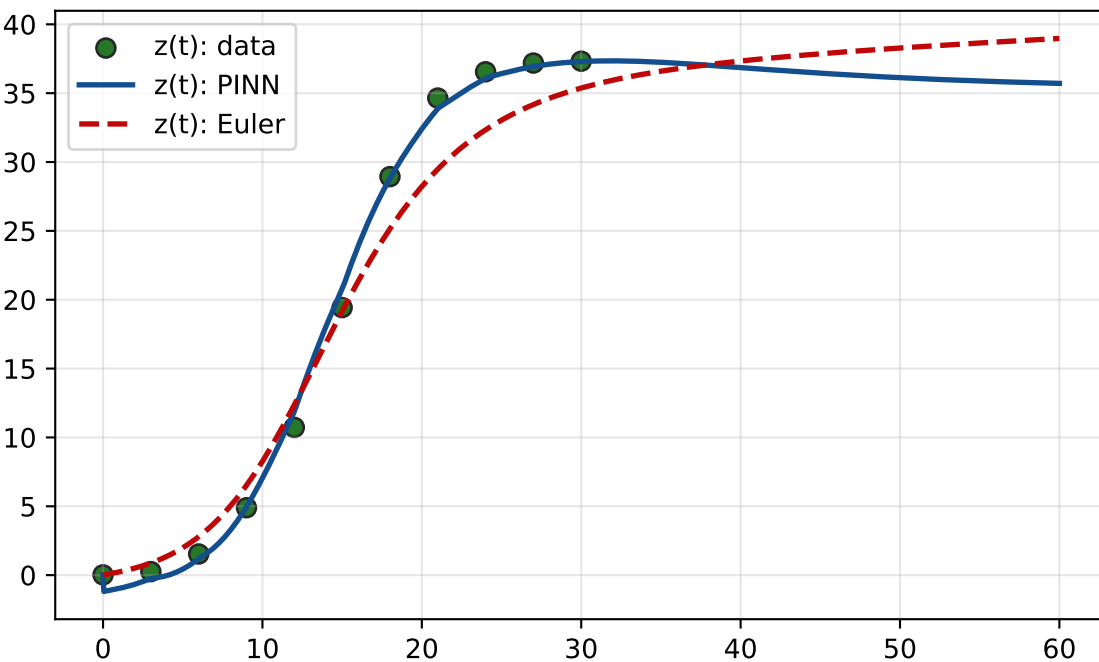
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E8

R2=0.9760, R2\_adj=0.9714, RSS=4.521e+02, AIC=1.196e+02

cond(JTJ)=1.23e+07, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 4.017935e-03 | 1.714061e-03 | 3.473017e-03 | 5.449177e-04  | 7.490953e-03 | 2.455191e-02 |
| p2    | 9.997330e-02 | 3.416532e-01 | 6.922552e-01 | -5.922819e-01 | 7.922285e-01 | 7.714502e-01 |
| p3    | 2.442150e-04 | 4.292339e-03 | 8.697105e-03 | -8.452890e-03 | 8.941320e-03 | 9.549345e-01 |
| p4    | 3.066250e-01 | 3.924263e-01 | 7.951313e-01 | -4.885063e-01 | 1.101756e+00 | 4.395639e-01 |
| p6    | 1.628527e-01 | 1.247067e-01 | 2.526798e-01 | -8.982711e-02 | 4.155326e-01 | 1.996526e-01 |
| p8    | 8.163717e-02 | 6.346828e-02 | 1.285989e-01 | -4.696177e-02 | 2.102361e-01 | 2.063407e-01 |
| p9    | 1.556469e-05 | 1.496296e-02 | 3.031784e-02 | -3.030227e-02 | 3.033340e-02 | 9.991756e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 2.993610e-02 |
| x0 (fixed)     | 5.180769e+00 |
| y0 (fixed)     | 6.814856e+01 |
| z0 (fixed)     | 1.921567e-02 |
| W0 (fixed)     | 0.000000e+00 |